

Quiz 1 Answer Key

Wednesday, March 4, 2026 5:39 PM

Question 1:

Which of the following best describes the Internet?

- ☐ A network of networks based on packet switching.
- ☐ A set of servers connected via HTTP.
- ☐ A network of networks based on circuit switching.
- ☐ A centralized network managed by the World Wide Web (WWW) organization

Question 2:

Two hosts are connected through a router. Link capacities are C1 and C2. What is the maximum achievable average throughput between Host1 and Host2?

Host1 --- C1 --- Router --- C2 --- Host2

- ☐ $\min\{C1, C2\}$
- ☐ $C1 + C2$
- ☐ $C1 * C2$
- ☐ $\max\{C1, C2\}$

Question 3:

Circuit switching is best characterized by {check all true answers}:

- ☐ No connection setup.
- ☐ Sending packets independently.
- ☐ The de facto communication method in use on the Internet.
- ☐ Efficient resource sharing.
- ☐ Performance guarantees.

Question 4:

Host1 and Host2 are connected through the Internet. Is it TRUE that packets sent from Host1 and Host2 will always follow the same fixed network path?

- ☐ True
- ☐ False

Question 5:

End devices on the Internet are also referred as:

- ☐ Routers
- ☐ Hosts
- ☐ Links
- ☐ Switches

Question 6:

What happens when a router receives packets at a higher rate than it can forward them through its outgoing links?

- ☐ Queuing delay increases
- ☐ Transmission delay increases
- ☐ Processing delay increases
- ☐ Propagation delay increases

Question 7:

Packet switching is best characterized by (check all true answers):

- ☐ Sending packets independently

- ☐ Efficient resource sharing
- ☐ Performance guarantee
- ☐ No connection setup
- ☐ The de facto communication method in use on the Internet

Question 8:

Host1 and Host2 are connected through a store-and-forward packet switching router. Enumerate the main types of delay in this network and briefly discuss whether they are generally *random* or constant.

Host1 ---- Router ---- Host2

Answer:

- Propagation delay: constant/deterministic
- Transmission delay: fairly constant/deterministic
- Queuing Delay: random
- Processing delay: fairly constant/deterministic